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**CS-3361-DATA SCIENCE PROJECT**
**ACADEMIC YEAR 2024-2025**
**MINI PROJECT**
**VOICE ASSISTANT**
**BRANCH & YEAR: CSE & 2nd YEAR**

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**Staff in charge department** **Head of the department**

# voice assistant

## Abstract

This Python program implements a Voice Assistant designed to interact with users using voice commands. The assistant, named Jarvis, utilizes various libraries to provide a range of features, making it both functional and user-friendly. Key components include:
Speech Recognition and Text-to-Speech: Using the speech_recognition library for capturing voice commands and pyttsx3 for vocal responses.
Web Integration: The assistant can search Wikipedia, open popular websites like YouTube, Google, and social media platforms, and fetch the latest news using web scraping (requests and BeautifulSoup).
Personalization: Users can set their names, rename the assistant, and engage in casual conversations, creating a more personalized experience.
Utility Functions: It provides the current time, answers general knowledge questions via WolframAlpha API, and pauses listening upon request.
Graphical User Interface (GUI): Developed using tkinter, the GUI enables user-friendly interaction and allows users to start the assistant easily.

## Introduction

This Python program introduces an interactive Voice Assistant, named Jarvis, capable of understanding and responding to voice commands. Built using a combination of libraries, it integrates functionalities for speech recognition, text-to-speech conversion, web browsing, and automation, making it a versatile tool for daily tasks.

The assistant listens to user commands through a microphone, processes them using the speech_recognition library, and responds using pyttsx3 for voice synthesis. Users can interact with the assistant to perform various tasks, including searching Wikipedia, fetching the latest news, accessing social media platforms, and even answering general knowledge queries through the WolframAlpha API.

The program is enhanced with a Graphical User Interface (GUI) built using tkinter, providing an intuitive interface for users to activate the assistant and monitor its activities. Its ability to learn and adapt names and preferences makes it highly customizable and user-centric.

# key words

**wolframalpha, pyttsx3, random , speech_recognition , wikipedia, webbrowser, os,datetime, requests, from bs4 import BeautifulSoup, tkinter, from tkinter import ***

**This Python project is a Voice Assistant application that allows users to interact with their computer using voice commands. It can perform tasks like web searches, fetching news, opening websites, giving time updates, and much more.**

**Usage:**

**1. Setup:**

**Ensure you have the required libraries installed:**

pip install wolframalpha
pyttsx3 SpeechRecognition
wikipedia request
beautifulsoup4

Replace "API_ID" with your WolframAlpha API key.

**2. Running the Program:**

**Execute the script, and the GUI window will appear.**

Click the **"Start"** button to activate the voice assistant.

**3. Commands to Try:**

**Greeting:** "Good Morning," "How are you?"

**Search:** "What is Python?" or "Who is Albert Einstein?"

**Open Websites:** "Open Google," "Open YouTube."

**Fetch News:** "News."

**Wikipedia Search:** "Search Python on Wikipedia."

**Change Name:** "Change my name to John."

**Current Time:** "What time is it?"

**Pause Listening:** "Stop listening for 10 seconds."

**4. Exiting:**

Say **"Exit"** to close the application.

## Tools and Libraries:

- **import wolframalpha**
- **import pyttsx3**
- **import random**
- **import speech_recognition as sr**
- **import wikipedia**
- **import webbrowser**
- **import os**
- **import datetime**
- **import requests**
- **from bs4 import BeautifulSoup**
- **import tkinter**
- **from tkinter import ***
- **import shutil**
- **import time**

## implementation:

### 1.Google

```
import webbrowser
elif 'open google' in command or 'open to google' in command:
        speak("Opening Google\n")
        webbrowser.open("google.com")
```

**Discussion:**

**Google Command:** Opens google.com when words like "open Google" or just "Google" .

### 2.youtube

```
import webbrowser
elif 'open youtube' in command or 'open to  youtube' in command :
      speak("Here you go, the Youtube is opening\n")
      webbrowser.open("youtube.com")
```

**Discussion:**

**YouTube Command:** Opens youtube.com when words like "open YouTube" or just "YouTube" are detected.

### 3.WhatsApp

```
import webbrowser
elif 'open whatsapp' in command or 'open to whatsapp' in command :
      speak("opening whatsapp\n")
      webbrowser.open("whatsapp.com")
```

**Discussion:**

**WhatsApp Command:** Opens whatsapp.com when words like "open whatsapp" or just "whatsapp" are detected.

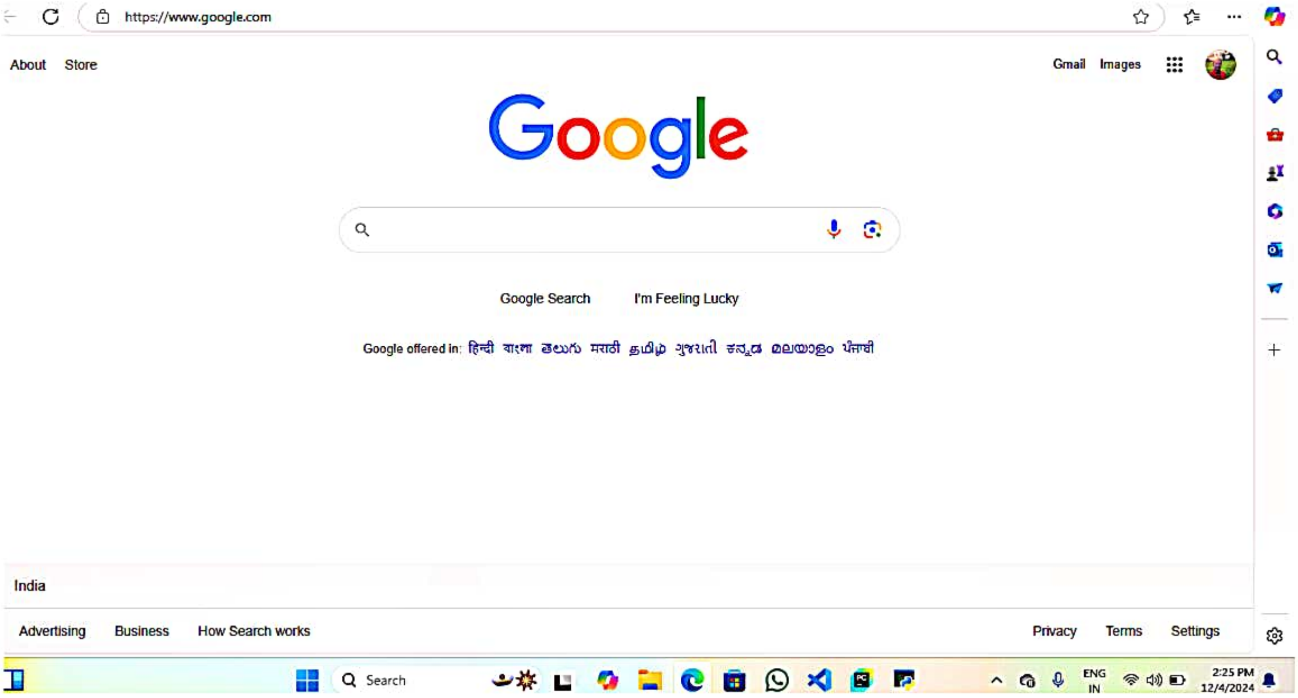
## style and design :

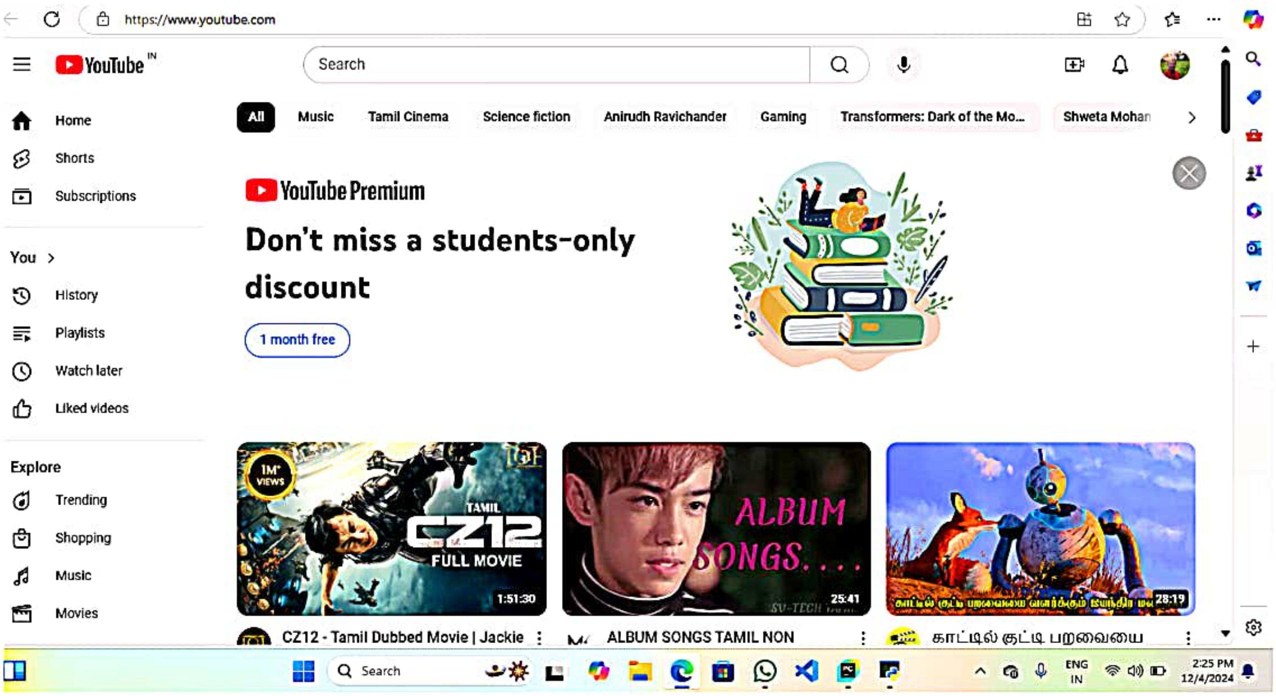
**This program serves as a basic GUI framework for interacting with a voice assistant. The functionality of callVoiceAssistant and how commands are updated in showCommand would need to be implemented for full functionality.**

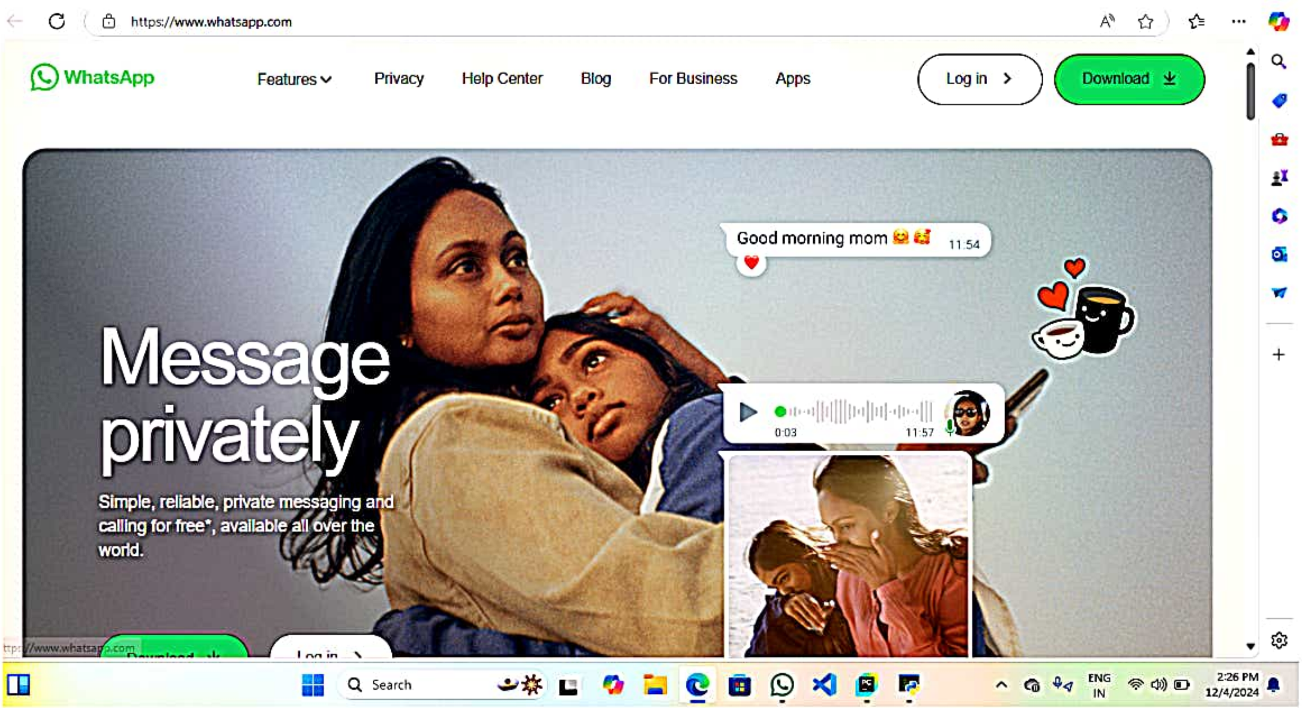
```
import tkinter
 #Creating the main window
wn = tkinter.Tk()
wn.title("Voice Assistant")
wn.geometry('700x700')
wn.config(bg='LightBlue1')

Label(wn, text='Welcome to meet the Voice Assistant ', bg='LightBlue1',
    fg='black', font=('Courier', 15)).place(x=50, y=10)

#Button to convert PDF to Audio form
Button(wn, text="Start", bg='gray',font=('Courier', 15),
     command=callVoiceAssistant).place(x=290, y=100)

showCommand=StringVar()
cmdLabel=Label(wn, textvariable=showCommand, bg='LightBlue1',
    fg='black', font=('Courier', 15))
cmdLabel.place(x=100, y=550)

#Runs the window till it is closed
wn.mainloop()
```

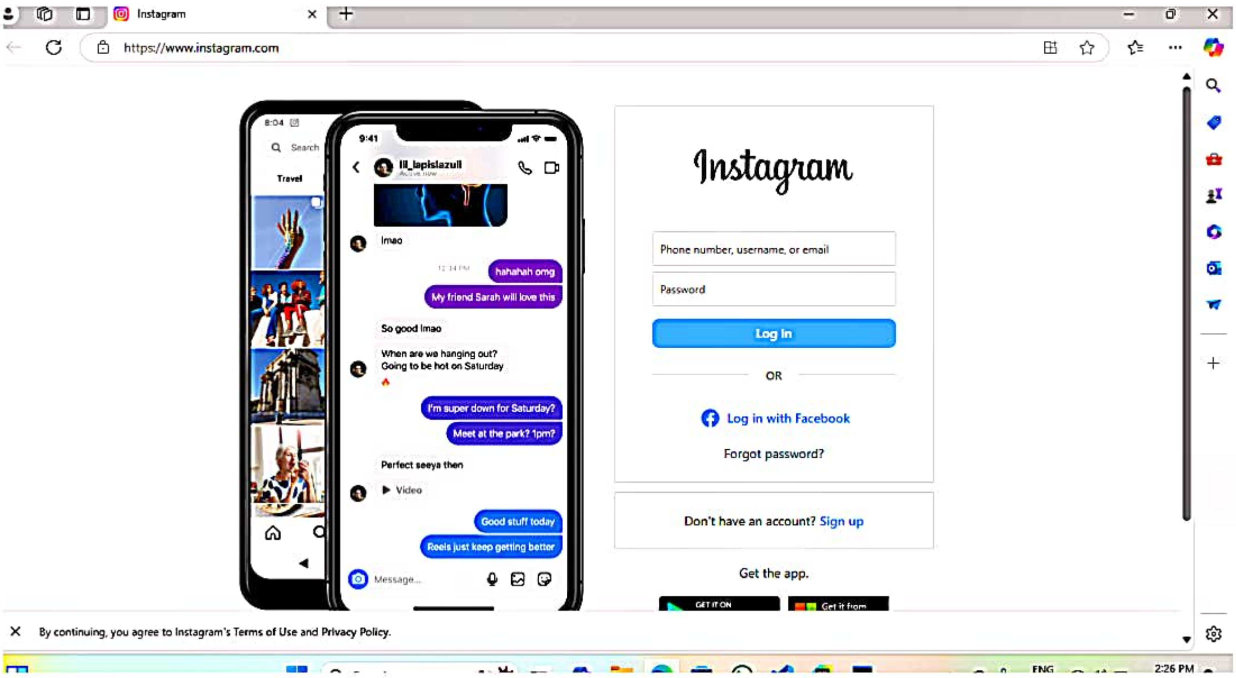
**tkinter is give a design and style of given python code**

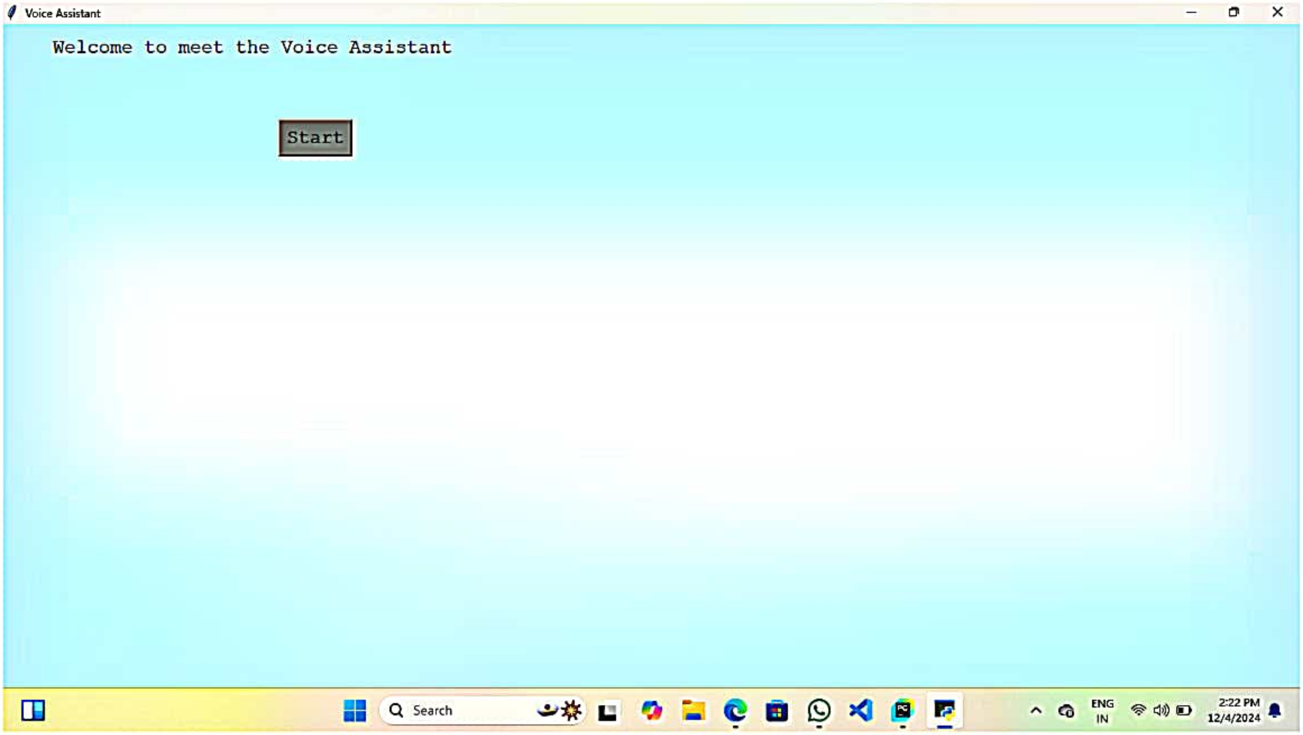
**Output:**

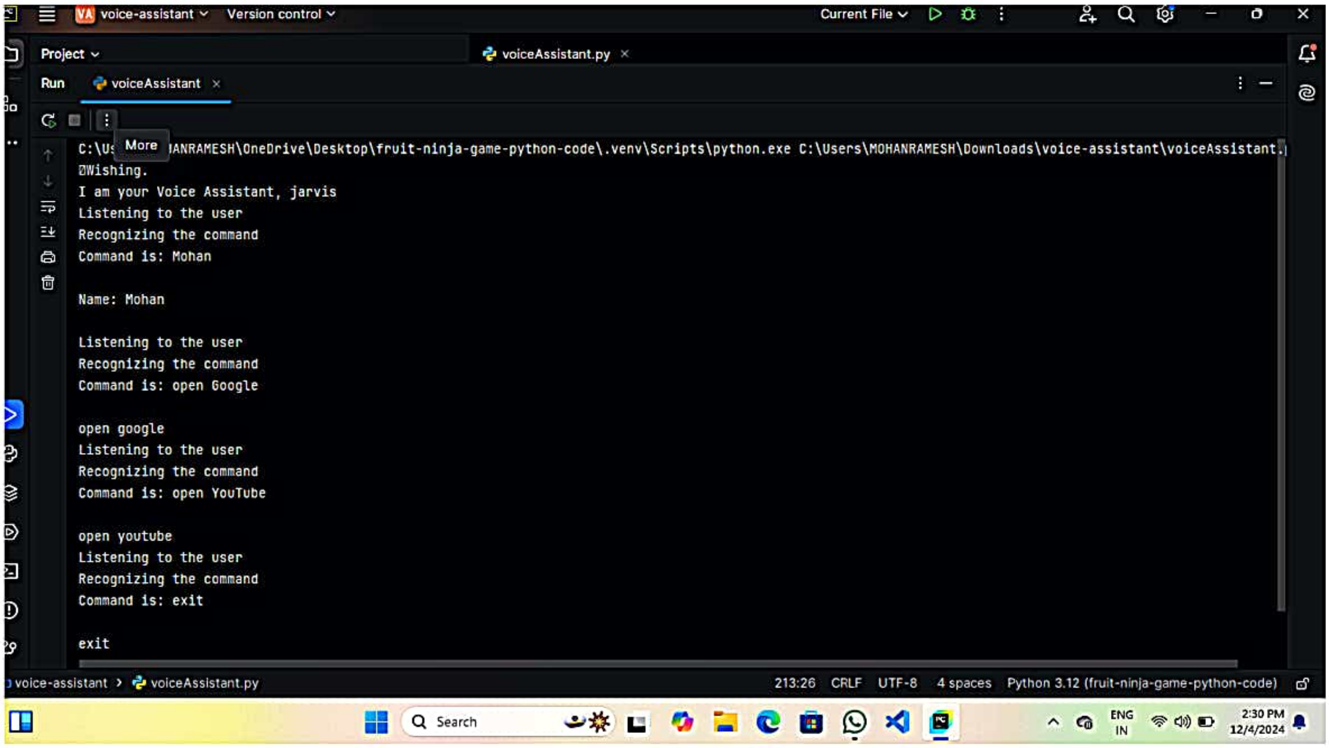

https://www.google.com
About
Store
Gmail
Images
Google
Google Search
I'm Feeling Lucky
Google offered in: हिन्दी বাংলা తెలుగు मराठी தமிழ் ગુજરાતી ಕನ್ನಡ മലയാളം ਪੰਜਾਬੀ
India
Advertising
Business
How Search works
Privacy
Terms
Settings
Search
ENG IN
2:25 PM 12/4/2024


https://www.youtube.com
YouTube IN
Search
Home
Shorts
Subscriptions
You >
History
Playlists
Watch later
Liked videos
Explore
Trending
Shopping
Music
Movies
All
Music
Tamil Cinema
Science fiction
Anirudh Ravichander
Gaming
Transformers: Dark of the Mo...
Shweta Mohan
YouTube Premium
Don't miss a students-only discount
1 month free
TAMIL CZ12 FULL MOVIE
1:51:30
ALBUM SONGS....
25:41
28:19
CZ12 - Tamil Dubbed Movie | Jackie
ALBUM SONGS TAMIL NON
காட்டில் குட்டி பறவையை
Search
ENG IN
2:25 PM 12/4/2024


https://www.whatsapp.com
WhatsApp
Features
Privacy
Help Center
Blog
For Business
Apps
Log in
Download
Good morning mom
11:54
0:03
11:57
Message privately
Simple, reliable, private messaging and calling for free*, available all over the world.
Search
ENG IN
2:26 PM
12/4/2024


Instagram
https://www.instagram.com
Instagram
Phone number, username, or email
Password
Log in
OR
Log in with Facebook
Forgot password?
Don't have an account? Sign up
Get the app.
By continuing, you agree to Instagram's Terms of Use and Privacy Policy.


Voice Assistant
Welcome to meet the Voice Assistant
Start
Search
ENG
IN
2:22 PM
12/4/2024


voice-assistant
Version control
Current File
Project
voiceAssistant.py
Run
voiceAssistant
More
C:\Us ANRAMESH\OneDrive\Desktop\fruit-ninja-game-python-code\.venv\Scripts\python.exe C:\Users\MOHANRAMESH\Downloads\voice-assistant\voiceAssistant.
Wishing.
I am your Voice Assistant, jarvis
Listening to the user
Recognizing the command
Command is: Mohan
Name: Mohan
Listening to the user
Recognizing the command
Command is: open Google
open google
Listening to the user
Recognizing the command
Command is: open YouTube
open youtube
Listening to the user
Recognizing the command
Command is: exit
exit
voice-assistant > voiceAssistant.py
213:26
CRLF
UTF-8
4 spaces
Python 3.12 (fruit-ninja-game-python-code)
Search
ENG
IN
2:30 PM
12/4/2024

## Conclusion:

This paper demonstrates the implementation of a voice assistant with a graphical user interface (GUI). It utilizes multiple libraries like pyttsx3, speech_recognition, wikipedia, and wolframalpha, among others, to provide various functionalities

## reference :

https://data-flair.com